



CS6P05ES Project

Interim Report

**TriplnMind: Personalized Travel Destination
Recommendation System**

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Declaration

Module: CS6P05ES

Deadline: 25/01/2026

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Abstract

Trip planning often involves navigating large volumes of unstructured information across multiple platforms, making the decision-making process time-consuming and inefficient. The aim of this project is to design and develop TripInMind, a personalized travel destination recommendation system that assists users in identifying suitable travel destinations based on individual preferences, constraints, and interests.

It involves the collection of structured user inputs such as preferred activities, travel duration, climate preference, and group type, and matching them with a structured destination database. A rule-based and preference matching technique is applied in producing travel recommendations. User account management, destination browsing, and recommendation history functionality is also provided in this system.

In this interim report, the project background, the activities accomplished to date, the development to be accomplished, as well as the project progress assessment in relation to the initial project plan are discussed. It is evident that the project is being pursued in an organized way and is on schedule to be completed on time.

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1. Introduction

This report documents the interim progress of the TripInMind project, a personalized destination recommendation system developed as part of the CS6P05ES module. This project has been developed to address key challenges faced by travellers when planning a trip by providing suitable destination recommendations based on user preferences.

The remainder of this report is structured as follows. Section 2 provides the background and contextual review of related systems and technologies. Section 3 presents the requirements analysis and specification. Section 4 describes the proposed solution design. Section 5 outlines the work completed to date. Section 6 discusses the remaining work required to complete the project, while Section 7 reviews the overall project progress in relation to the planned schedule. Section 8 presents the project management aspects, including planning and scheduling. Finally, Sections 9 and 10 present the references and bibliography used during development.

1.1 Project Scope and Objectives

The primary objective of the project is to design and implement a web-based system that recommends travel destinations based on user-defined criteria. The system is intended to support travellers by reducing planning complexity and improving decision accuracy through personalized suggestions.

2. Background

The rapid rise of online tourism platforms has further increased the requirement for intelligent travel recommendation systems that have to serve users with greater efficiency while planning their tour. Online travel platforms, such as TripAdvisor, generally offer users with various options based on popularity and promotion. However, these platforms provide users with abundant information, yet such platforms are generally incapable of providing users with adequate recommendations based on their preferences and needs, such as activity interests, travel duration, and group types. Users are thus often forced to evaluate various options, making the overall experience inefficient [1], [2].

Past studies in the domain of recommendation systems have identified various techniques to improve the personalization in the system. Rule-based recommendation systems operate by using rules to map the user preferences to the options provided. They ensure transparency in the operation. Moreover, content-based recommendation techniques operate by using the user inputs to ensure the options provided to the user correspond to the user's interests. According to various studies, the integration of the preference filtering mechanism with the rules would improve the accuracy of the recommendation system in the domain of travel planning due to the variability in the user requirements [1], [2].

The development of travel recommendation systems heavily depends on the development of web-based information systems. Database-based information systems are often used to effectively store and retrieve destination information, user information, and user preferences. Also, in the development of the front end of the information systems, technologies like HTML and CSS may often be used to develop user-friendly interfaces. Additionally, PHP and MySQL may often be used in the development of information systems. Also, human-computer interaction plays an important role in the development of information systems [3], [4].

Software quality factors have a major influence in the development of such software. Various international standards such as ISO/IEC 25010 have emphasized the importance of various quality factors such as usability, reliability, and maintainability in software. These factors have great significance in the case of web-based recommendation systems. Considering the quality factors along with the existing techniques in the area of recommendation systems, the project named “TripInMind” strives to address the shortcomings in the existing web platforms [5].

3. Requirements Analysis and Specifications

3.1 Problem Definition

The rapid growth in online travel websites has led to an information explosion for users looking to consume information online regarding their travel plans. While currently online websites offer a plethora of information regarding destinations, they are mostly dependent upon traditional methods of recommending destinations or relying upon popular destinations or sponsored content, which may not take into account the various personalization factors of the users, such as desired activities to be performed at the destinations, length of visit, desired climatic conditions, or types of groups visiting the destinations. As a result, it becomes a complex process for the users to evaluate various destinations manually.

The fundamental question addressed through this project is the lack of an organized and personalized mechanism to help users find the right travel destinations based on their specific needs and requirements. TripInMind is designed to reduce this problem through the use of user preferences to generate recommendations on appropriate travel destinations.

3.2 Functional Requirements

The functional requirements specify the capabilities that must be enabled within the TripInMind system to meet its objectives. The primary functional requirements that have been identified for this system are as follows:

- The system shall allow users to register and have an account created.
- The system will allow users who are already registered to log in.
- The system will be responsible for collecting user preferences such as duration of travel, activity preferences, climatic preferences, and types of groups.
- User information and preferences shall be stored in a structured database.
- The system should provide recommendations regarding the destination of a trip based on the provided information regarding the user's preferences.
- The system shall display recommended destinations along with relevant information to the user.
- The system will allow users to view the list of available destinations without the need to log in.

3.3 Non-functional Requirements

The non-functional requirements describe the quality attributes and constraints that affect the system. The system developed with TripInMind has the following non-functional requirements:

1. Usability: The system shall offer an interface that is simple, intuitive, and user-friendly to accommodate users from various technical backgrounds.
2. Performance: The system shall be able to generate recommendations for travel destinations within an acceptable response time.
3. Reliability: The system shall operate reliably, and accurate recommendations shall be produced without loss of data.
4. Security: In the case of user authentication data, the information should be kept secure to prevent unauthorized access.
5. Scalability: The system will be able to support an increasing number of users and/or destinations without compromising the quality of the system.
6. Maintainability: The system will be designed with a modular architecture to make it easier to upgrade or enhance it in the future.

4. SOLUTION DESIGN

In this section, the overall architecture of the TripInMind system will be described, as well as the justification of the architectural approach followed for the system's design in relation to the proposed logic for the generation of recommendations and data management.

4.1 System architecture Overview

The architecture of the TripInMind system is based on a client-server model with a web-based interface. HTML and CSS are used to implement the client side of the system to provide a user-friendly interface. The server side of the system is implemented in PHP to handle user authentication, processing of user preference data, and the actual recommendation process. A MySQL database is used to store data on various destinations as well as user data.

This design choice is based on considerations of simplicity and suitability for the purpose. Also, the design is geared towards the presentation layer, the logic layer, and the data layer.

4.2 Recommendation Logic Design

The recommendation process is carried out through a rule-based and preference-matching approach. This approach is selected as there are specific and distinct constraint sets in travel planning, including the travel duration set, activity set, climate set, and group set. The use of a rule-based approach offers transparency in the decision-making process.

The input provided to the system through its user interface is compared to the attributes of the desired destination, which have been stored in the database. A set of rules is used to determine the extent to which user preferences match available options for the desired destination. Those options that match the highest number of criteria are given preference and provided to the user as recommendations. This

ensures consistency, transparency, and simplicity while still allowing for personalization to be effective.

4.3 System Flow and Data Processing

The procedural flow starts when a user interacts with the TripInMind platform, thereby entering his or her travel-related preferences through the recommendation interface. The entered information is validated and processed by the application running on the server side. The processed information, as derived from the above step, is utilized to retrieve destination-related information from the destination database.

The logic of the recommendation system evaluates the retrieved destination based on the user's preference criteria defined by set rules. Consequently, a list of ranked travel destinations relevant to the user is created. Finally, the generated travel recommendations are provided to the user via the front-end interface with relevant destination information. For users who are registered with the system, the results of the travel recommendations can also be saved for future use.

This structured approach helps in efficient data processing, recommendation generation, and providing an unbroken experience.

4.4 Database Design and Personalization Support

The database plays a major part in facilitating personalization in the TripInMind system. Information about destinations with regard to aspects like location, activities, weather, and suitability for different groups is stored in a database. Information about users is also stored in a database.

With the use of relation tables for both user and destination information, the system is able to access relevant information during the recommendation process through efficient and scalable design, which is beneficial for any future enhancements to the system, including filtering and feedback from users.

5. Work Completed

At the interim stage, some important activities have been effectively accomplished within the context of the TripInMind project. The activities include problem domain analysis and system requirements specification, including both functional and non-functional specifications. Following the specification of system requirements, overall system architecture and database schema have been designed by utilizing Entity-Relationship Diagrams (ERDs) for data management.

In addition to this, wireframes for key interfaces of the system were developed. These include home page interface, login interface, registration interface, and recommendation interface. Front-end interfaces for user interaction were developed using HTML and CSS to lay out the basic structure of the system. Development of user authentication was conducted to lay out user access to the system. In addition to this, UML diagrams for use cases and structured architecture were developed to support system design.

From the finished work, it is clear that it follows a logical approach to system development, which is consistent with the expected outcomes of the supervision meetings that were held.

6. Further Work

Some fundamental activities need to be carried out to conclude the development of the TripInMind system. These include carrying out the destination recommendation logic, integrating the destination recommendation algorithm with the system database, and enabling destination recommendations for personalization. The development of other backend functionalities using PHP is also needed.

In addition, further development will place greater emphasis on user profile management and the secure storage of user preferences in order to improve the personalization features offered by the system. Thoroughly planned tests will be carried out in order to guarantee functional correctness and stability in the system. Performance tests will be conducted in order to determine the usability of the system.

Lastly, there is a set of remaining tasks, which include refining the system according to the feedback received from the tests, as well as generating the final project report. These remaining tasks can be completed within the allocated timeframe, and their completion is considered consistent with the initial project plan.

7. Progress Review

With regard to the interim juncture, it is noted that the TripInMind project is progressing largely in line with the planned schedule. The major activities, such as requirements analysis, system design, database modelling, and initial front-end development, have been satisfactorily concluded, which is consistent with the milestones identified in relation to the early stages of the project, thereby reflecting a progression towards the overall objectives of the project.

At various stages of development, some minor challenges were identified; however, these challenges were identified in relation to server configuration. Such challenges are likely to lead to some delay in development; nevertheless, different approaches are used to ensure that development proceeds. Therefore, it can be noted that the challenges identified did not greatly impact the development schedule.

It is evident that the original project plan is still valid, with minor adjustments being made to the project's implementation timeline due to technical challenges. The outstanding project activities, including implementation, testing, and documentation, have been reviewed and will be carried out within subsequent project phases. Therefore, from the progress made so far, it is evident that the project is on track to be completed within the specified timeframe.

8. Project Management

Effective project management is key in guaranteeing that the project of developing the TripInMind system is completed in a timely fashion as desired. A project plan was formulated, which helped in coordinating various project activities through various project phases.

The project timeline includes the whole project lifecycle from the planning phase to the final submission phase. The work done so far includes the analysis of the requirements, system design, creation of the wire frame, and initial front-end work. Future work includes the implementation of the recommendation logic and the final testing phase.

Project progress is tracked through the establishment of project milestones to ensure proper alignment with the project schedule. Project milestones include the submission of the interim report and the submission of the final system. Overall project schedule and task allocation have been provided through the utilization of a Gantt chart.

9. References

- [1] I. Sommerville, Software Engineering, 10th ed. Boston, MA, USA: Pearson Education, 2016.
- [2] R. S. Pressman, Software Engineering: A Practitioner's Approach, 8th ed. New York, NY, USA: McGraw-Hill Education, 2014.
- [3] W3Schools, "HTML, CSS and PHP Documentation," 2024. [Online]. Available: <https://www.w3schools.com>

10. Bibliography

[4] Mozilla Developer Network, “Web Development Documentation.” [Online]. Available: <https://developer.mozilla.org>

[5] ISO/IEC, ISO/IEC 25010: Systems and Software Quality Requirements and Evaluation. Geneva, Switzerland: ISO, 2011.

11. Appendix

Appendix A: System Diagrams



Figure A 1 :Entity Relationship Diagram (ERD) of the TripInMind System

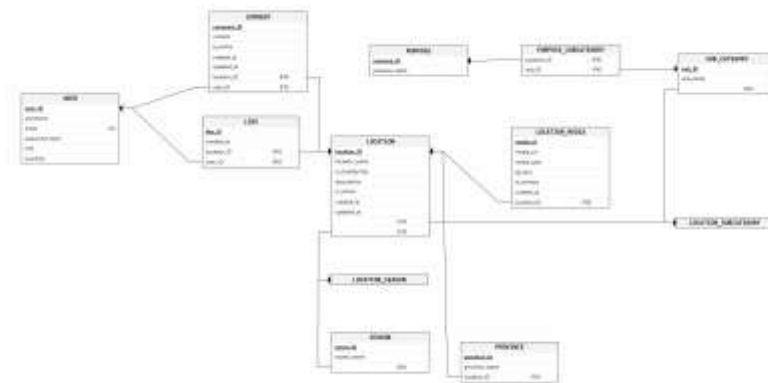


Figure A 2: Conceptual ER Model Showing Entities, Attributes, and Relationships

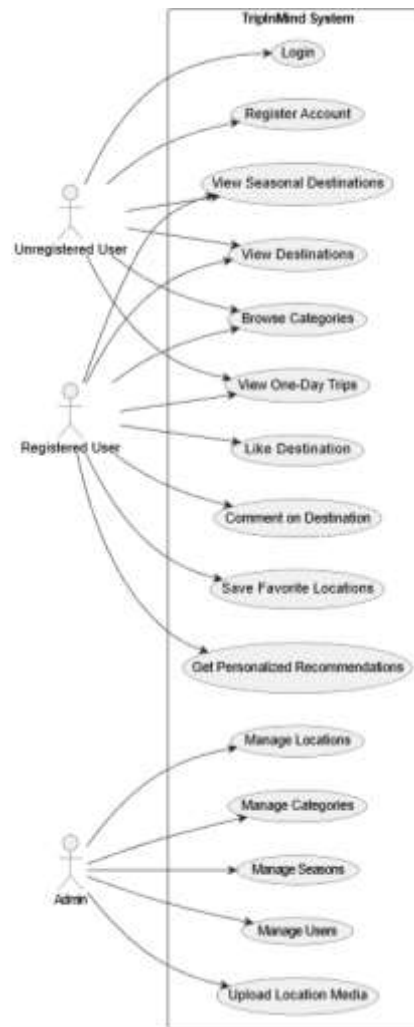


Figure A 3: Use Case Diagram of the TripInMind System

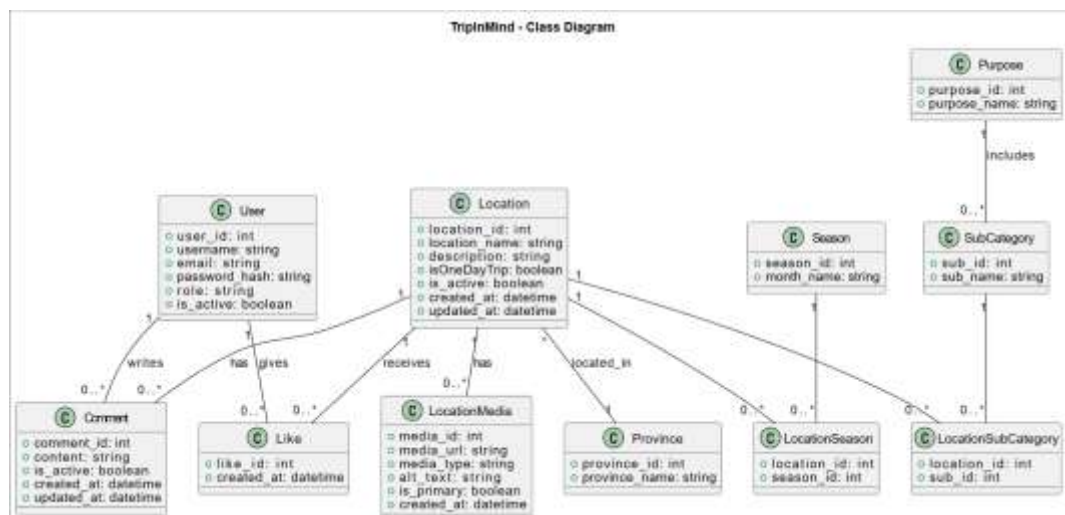


Figure A 4: Class Diagram

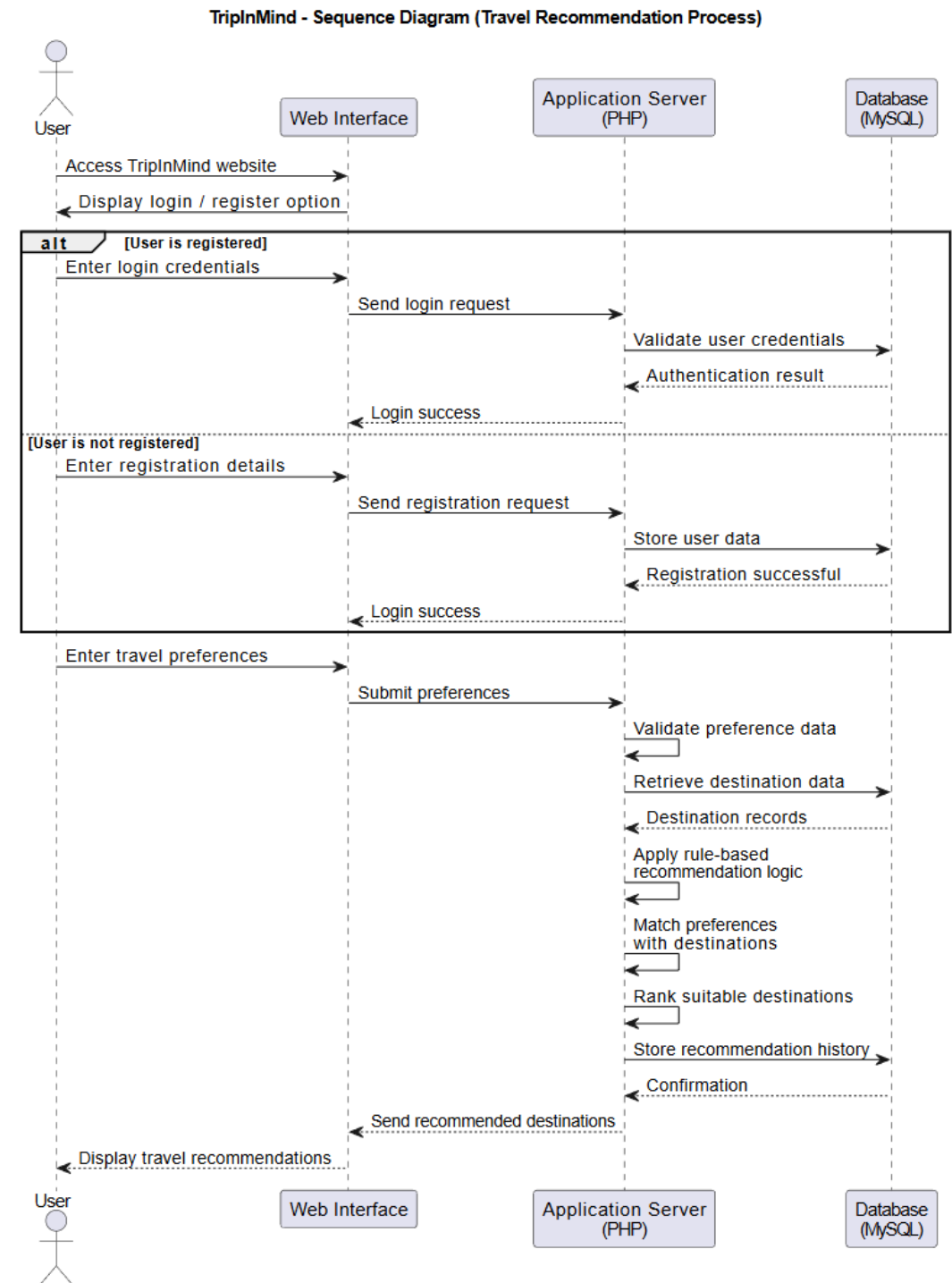


Figure A 5: Sequence Diagram

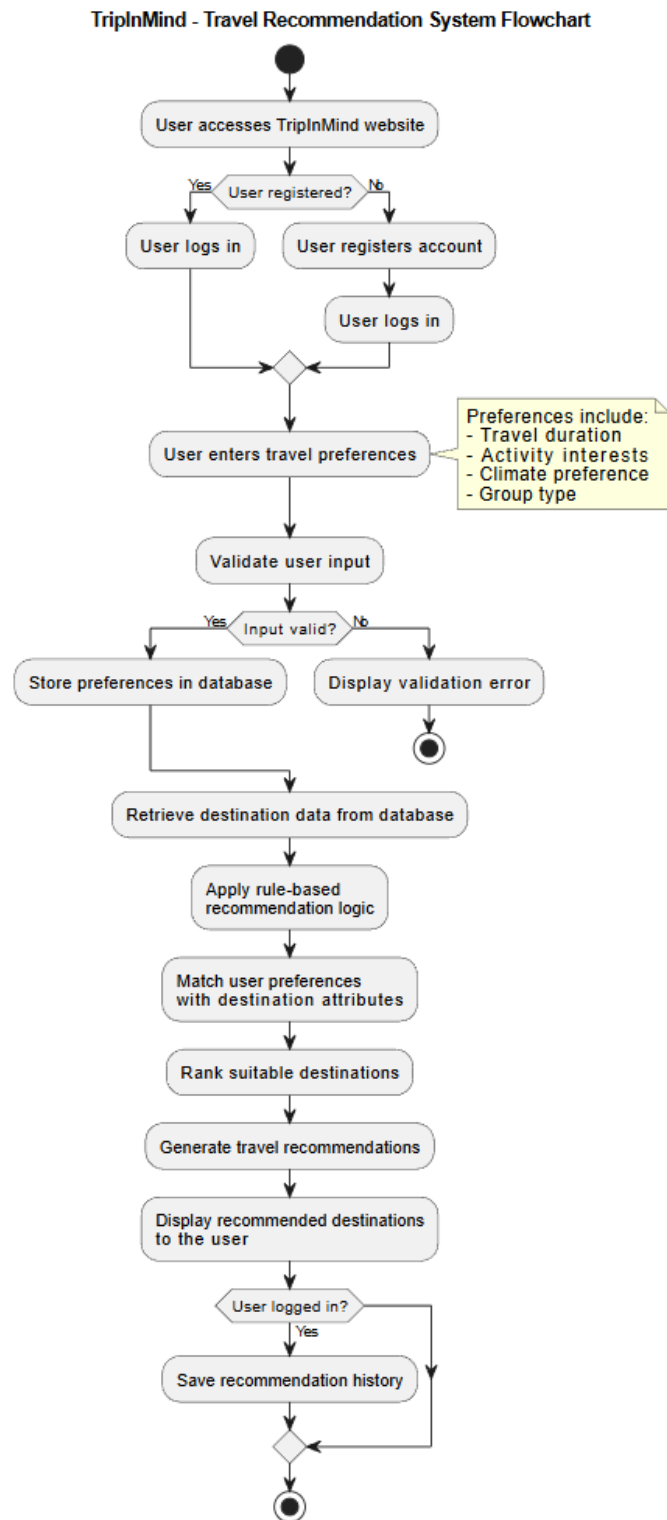
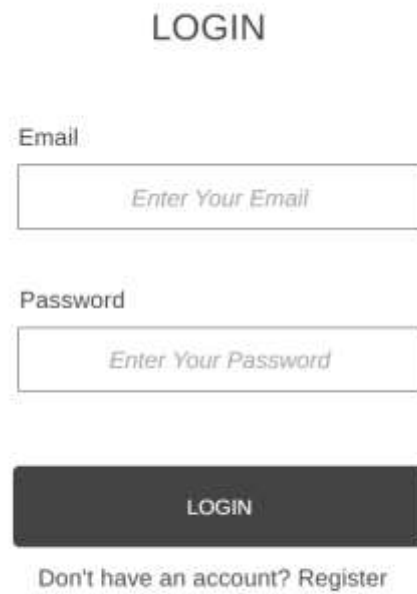


Figure A 7: System Flowchart

Appendix B: User Interface Wireframes / Interfaces



A login page wireframe with a centered title 'LOGIN'. Below the title are two input fields: 'Email' and 'Password'. Each field has a placeholder text 'Enter Your Email' and 'Enter Your Password' respectively. Below the input fields is a dark grey button labeled 'LOGIN'. At the bottom, there is a link 'Don't have an account? Register'.

LOGIN

Email

Enter Your Email

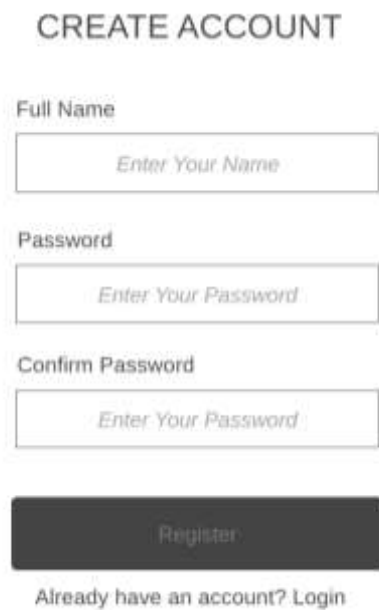
Password

Enter Your Password

LOGIN

Don't have an account? Register

Figure B 1: Login Page Wireframe



A register page wireframe with a centered title 'CREATE ACCOUNT'. Below the title are three input fields: 'Full Name', 'Password', and 'Confirm Password'. Each field has a placeholder text 'Enter Your Name', 'Enter Your Password', and 'Enter Your Password' respectively. Below the input fields is a dark grey button labeled 'Register'. At the bottom, there is a link 'Already have an account? Login'.

CREATE ACCOUNT

Full Name

Enter Your Name

Password

Enter Your Password

Confirm Password

Enter Your Password

Register

Already have an account? Login

Figure B 2: Register Page Wireframe

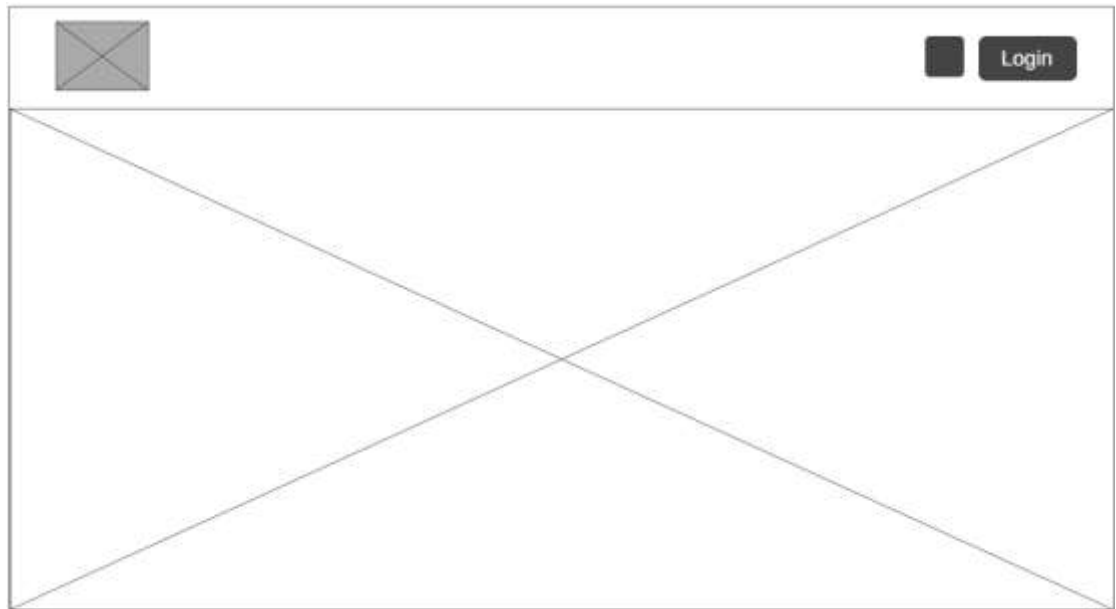


Figure B 3: Home Page Wireframe

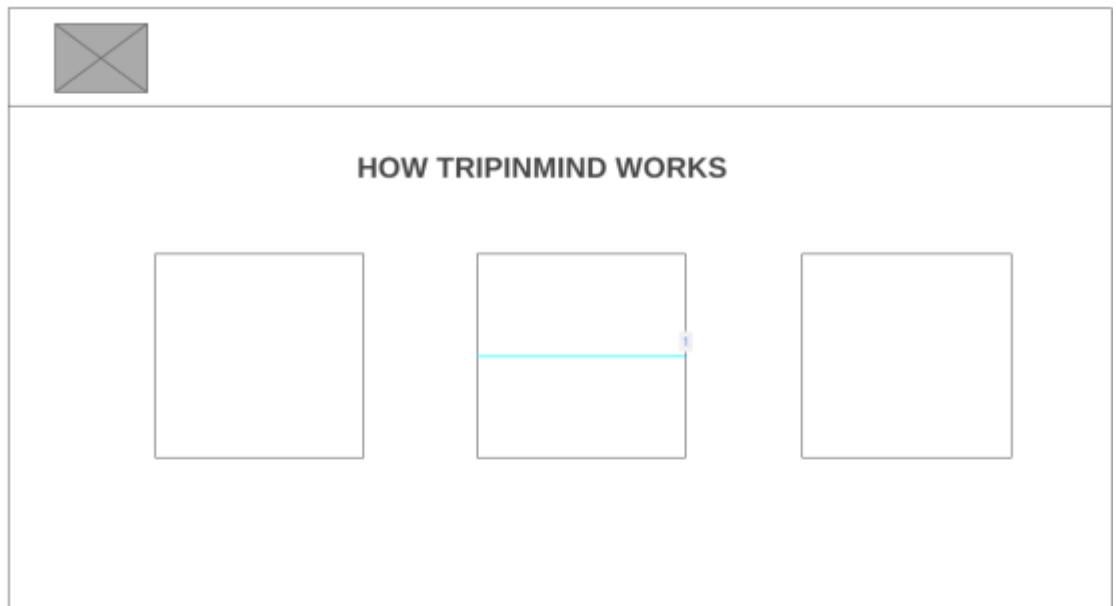


Figure B 4: Page 01 Wireframe



Figure B 5: Categories Page Wireframe

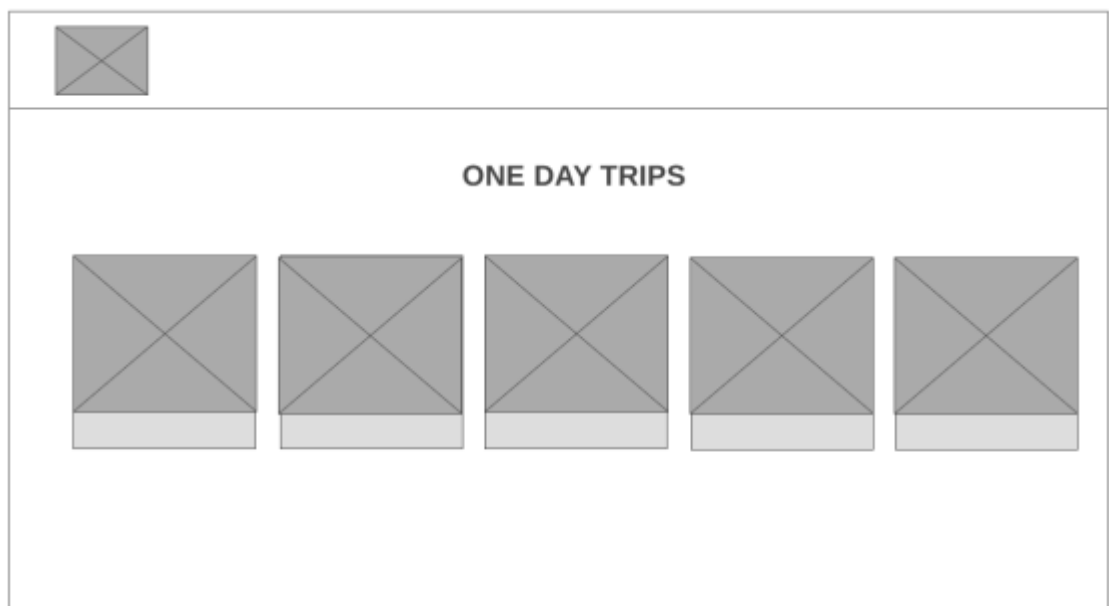


Figure B 6: One Daytrip Page Wireframe



Figure B 7: Popular Destinations Page Wireframe



Figure B 8: Recommendations Page Wireframe

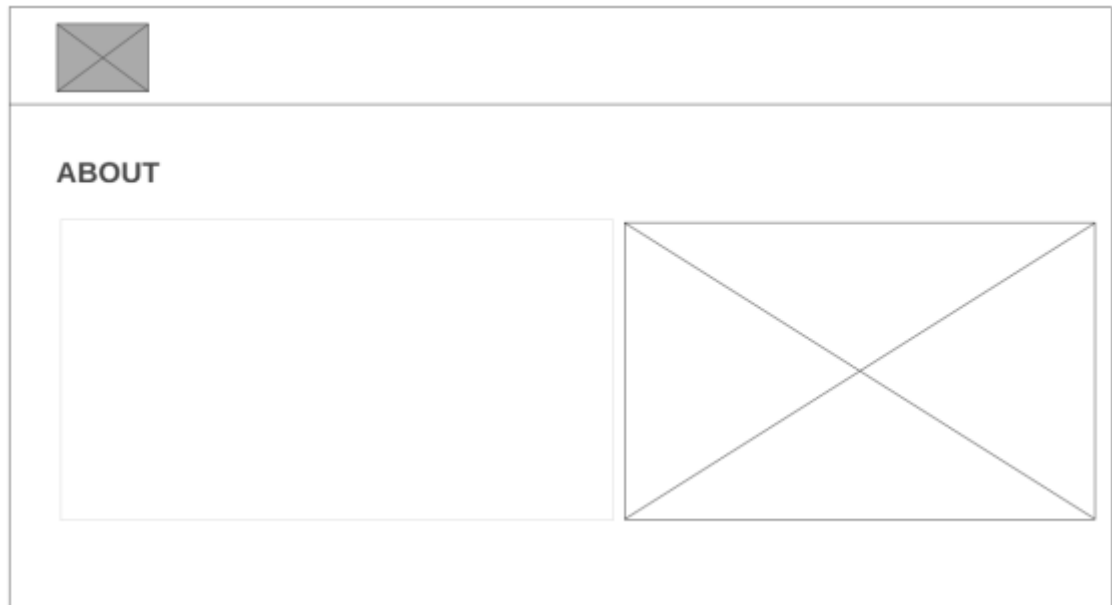
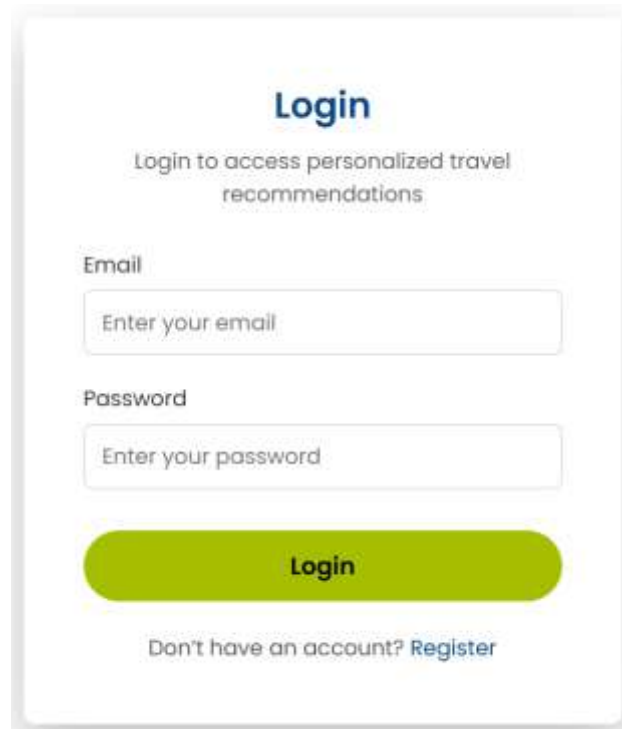


Figure B 9: About Page Wireframe

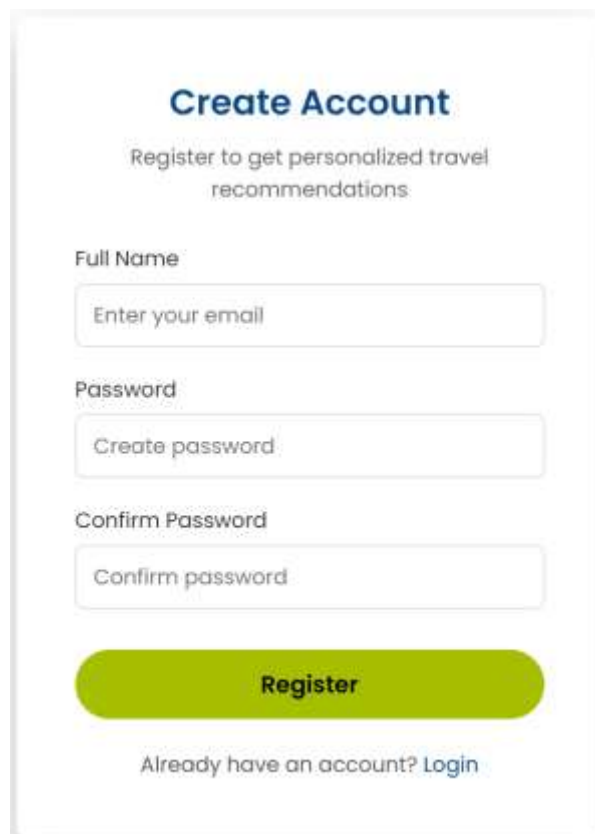


Figure B 10: Admin Dashboard Wireframe



The login page interface features a white background with a light gray border. At the top, the word "Login" is displayed in a bold, dark blue font. Below it, a subtitle reads "Login to access personalized travel recommendations". The form includes two input fields: "Email" with the placeholder text "Enter your email" and "Password" with the placeholder text "Enter your password". A prominent green "Login" button is centered below the fields. At the bottom, a link states "Don't have an account? [Register](#)".

Figure B 11: Login Page Interface



The register page interface has a white background with a light gray border. The title "Create Account" is at the top in a bold, dark blue font, followed by the subtitle "Register to get personalized travel recommendations". The form contains three input fields: "Full Name" with the placeholder "Enter your email" (likely a typo for "name"), "Password" with the placeholder "Create password", and "Confirm Password" with the placeholder "Confirm password". A green "Register" button is positioned below these fields. At the bottom, a link says "Already have an account? [Login](#)".

Figure B 12: Register Page Interface

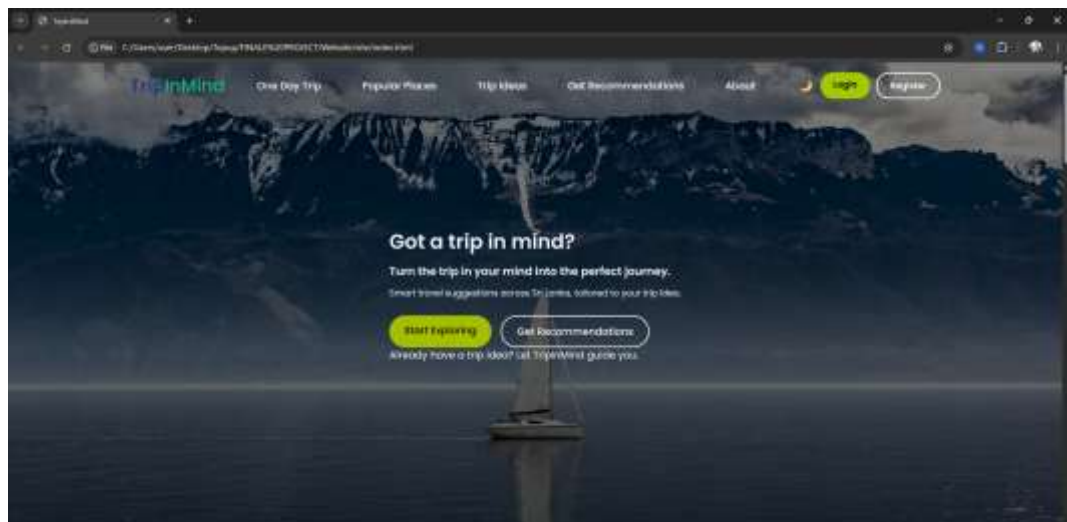


Figure B 13: Home Page Interface

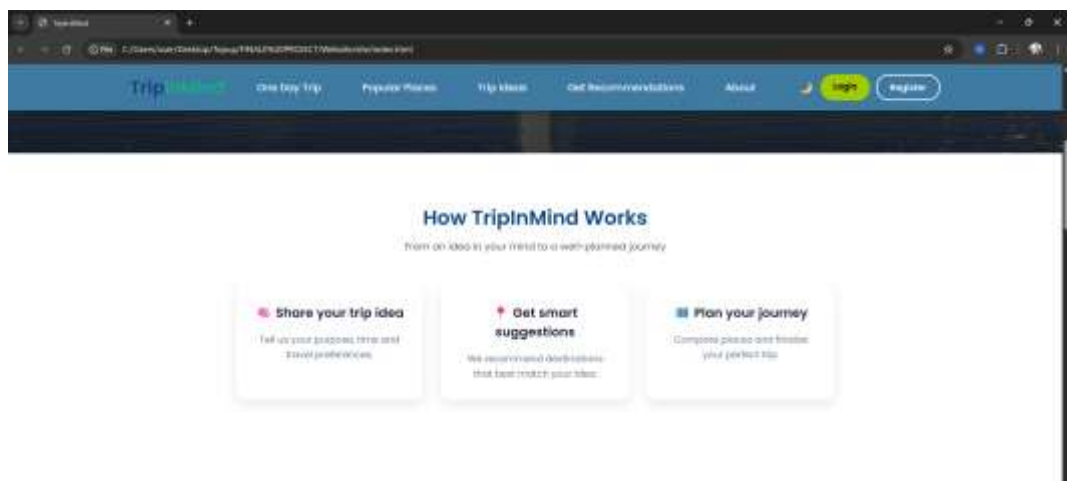


Figure B 14: Page 01 Interface

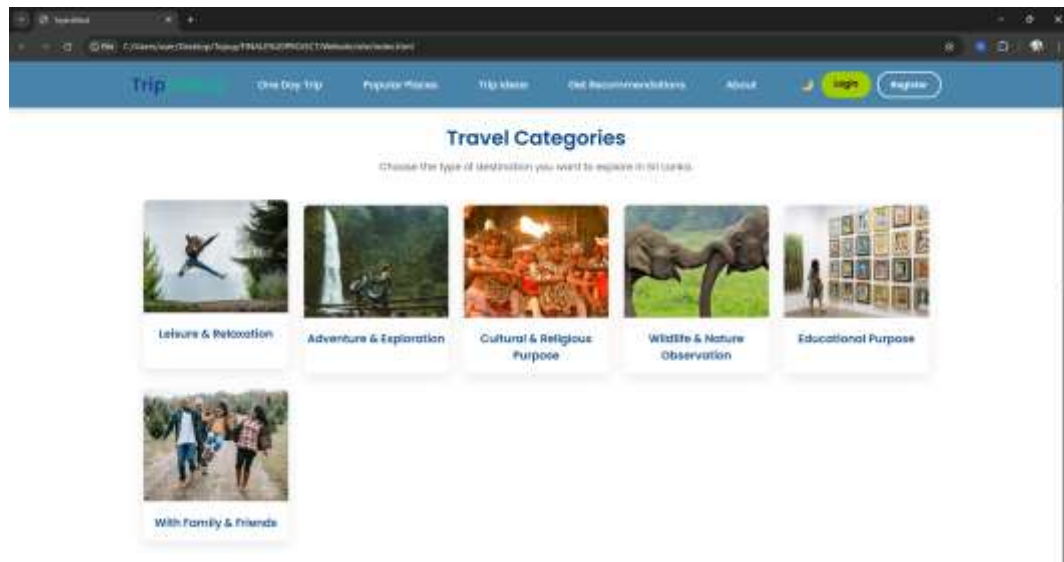


Figure B 15: Categories Page Interface

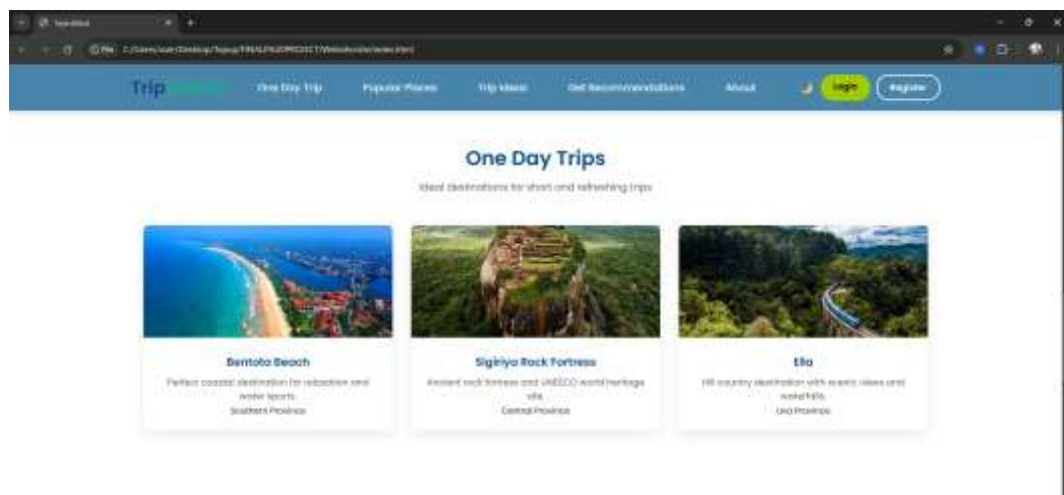


Figure B 16: One Day Trips Page Interface

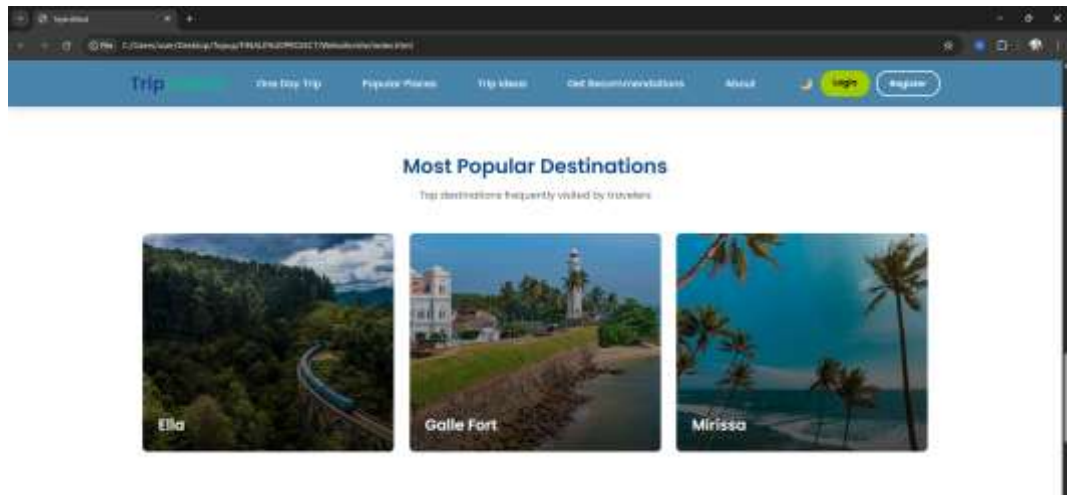


Figure B 17: Popular Destination Page Interface

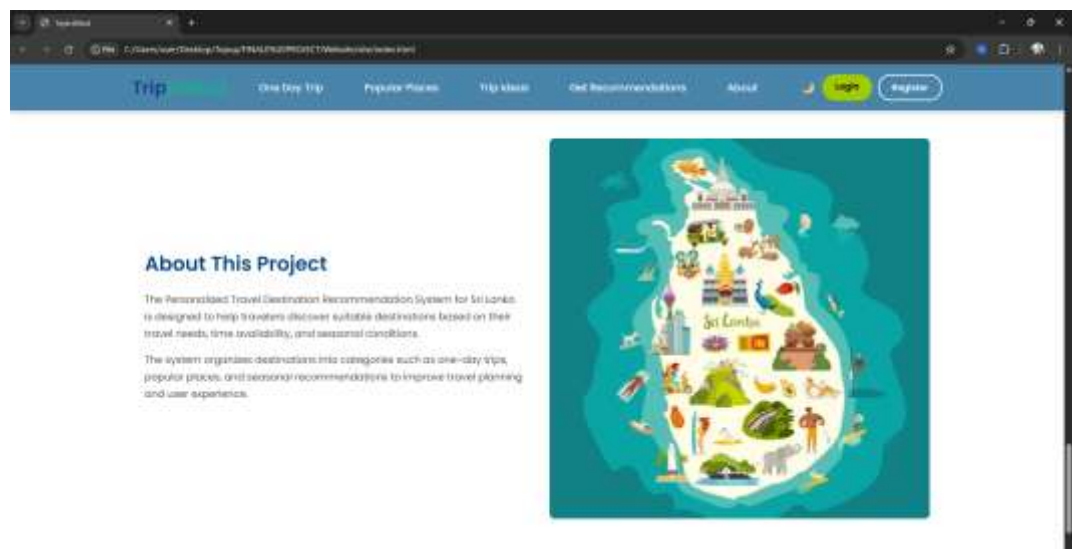


Figure B 18: About Page Interface



Figure B 19: Footer Interface

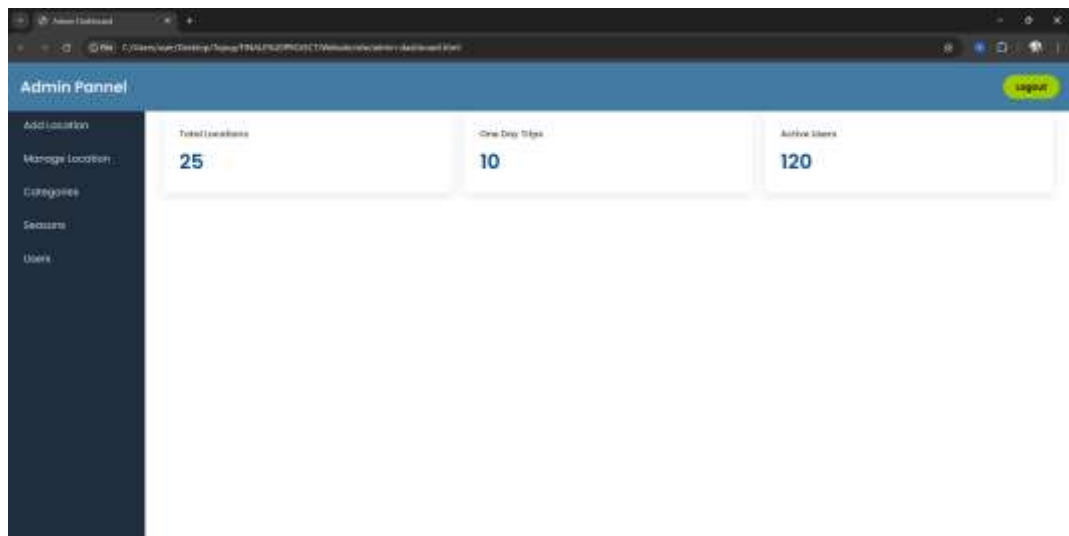


Figure B 20: Admin Dashboard

Appendix C: Project Management

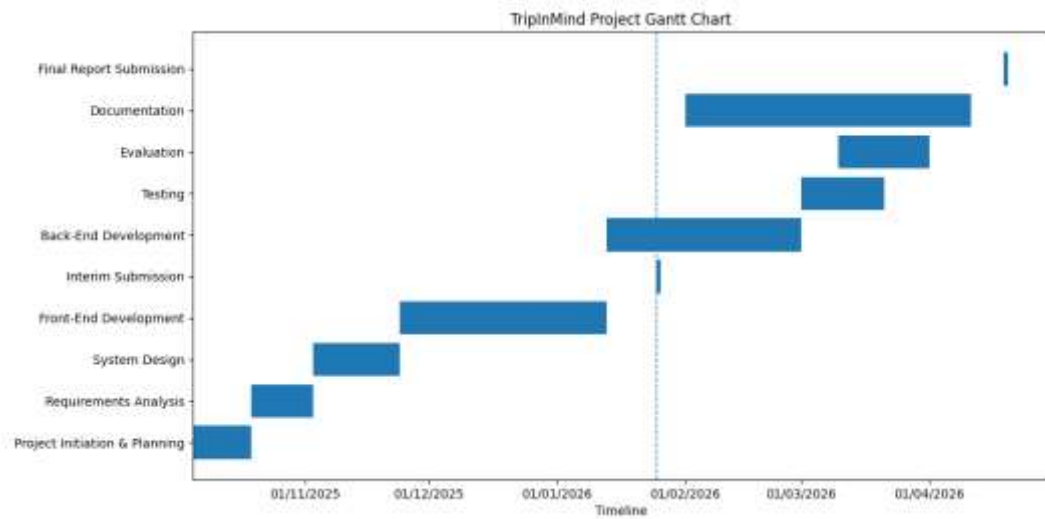


Figure C 1: Gantt Chart